

Buyback Dividend Preference and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Buyback Dividend Preference (BDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on BDP achieves an annualized gross (net) Sharpe ratio of 0.49 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 43 (35) bps/month with a t-statistic of 4.67 (3.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square) is 28 bps/month with a t-statistic of 3.05.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent framework for understanding risk premia, empirical identification of the channels through which firms' payout policies affect their consumption risk exposure remains incomplete. We examine how firms' choice between share buybacks and dividends—two economically equivalent methods of returning cash to shareholders under perfect markets—reveals information about their exposure to aggregate consumption risk and hence their required returns. This paper documents that Buyback Dividend Preference (BDP), measuring firms' relative propensity to distribute cash through repurchases rather than dividends, strongly predicts cross-sectional variation in expected returns, with firms exhibiting higher buyback preference earning significantly higher risk-adjusted returns.

The predictive power of buyback dividend preference for returns emerges through its connection to firms' exposure to long-run consumption risk and state-dependent marginal utility. Firms choosing buybacks over dividends signal greater flexibility in their payout policies, allowing them to adjust distributions more readily in response to aggregate consumption shocks [Jagannathan et al. \(2000\)](#). This flexibility becomes particularly valuable during economic downturns when investors' marginal utility of consumption is high, as firms can suspend buybacks without the negative signaling effects associated with dividend cuts [Grullon and Michaely \(2002\)](#). The differential information content and commitment implied by these payout methods affects how investors perceive firms' ability to smooth consumption across states, with buyback-intensive firms exhibiting higher sensitivity to consumption growth volatility [Boudoukh et al. \(2007\)](#).

From a consumption-based perspective, firms with high buyback preference load

more heavily on systematic risk factors that covary with the stochastic discount factor, particularly during periods of heightened consumption uncertainty. The optionality embedded in buyback programs exposes investors to greater consumption risk during bad states when the correlation between firm cash flows and aggregate consumption strengthens [Dittmar \(2000\)](#). This state-dependent exposure manifests through firms’ differential responses to disaster risk and macroeconomic volatility, as buyback-intensive firms experience larger valuation adjustments when investors update their beliefs about long-run consumption growth [Bansal and Yaron \(2004\)](#).

The intertemporal marginal rate of substitution prices the differential risk exposure between high and low BDP firms through multiple channels. First, buyback preference correlates with firms’ exposure to long-run risks in aggregate consumption, as flexible payout policies amplify the transmission of persistent consumption shocks to firm valuations [Hansen et al. \(2008\)](#). Second, the procyclical nature of buybacks relative to dividends creates time-varying risk exposure that commands a premium when investors exhibit habit formation preferences [Campbell and Cochrane \(1999\)](#). Third, the concentration of buyback activity among growth firms with volatile cash flows increases their sensitivity to innovations in the consumption-wealth ratio, requiring higher expected returns to compensate investors for bearing this systematic risk [Lettau and Ludvigson \(2001\)](#).

Our empirical analysis reveals that a value-weighted long-short strategy based on buyback dividend preference achieves an annualized Sharpe ratio of 0.49 before transaction costs, placing it in the 89th percentile among documented anomalies. The strategy generates average monthly returns of 34 basis points with a t-statistic of 3.58, demonstrating robust predictive power across our sample period from 1972 to 2024. Most notably, the BDP strategy maintains significant alpha of 43 basis points per month (t-statistic = 4.67) relative to the Fama-French six-factor model, indicating that existing risk factors do not fully capture the consumption risk exposure

associated with payout policy choices.

The economic magnitude of the BDP effect remains substantial after accounting for transaction costs, with net returns of 31 basis points per month and a net Sharpe ratio of 0.45 that ranks in the 97th percentile of anomalies. The strategy’s performance persists across different portfolio construction methodologies, with significant alphas ranging from 26 to 61 basis points depending on the sorting procedure and weighting scheme employed. Importantly, the predictive power of BDP extends to the largest stocks, generating returns of 36 basis points per month (t-statistic = 3.30) among firms above the 80th NYSE size percentile, alleviating concerns about small-stock effects driving our results.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the BDP strategy retains an alpha of 28 basis points per month with a t-statistic of 3.05, confirming that our signal captures unique variation in expected returns. The strategy loads significantly on the momentum factor with a coefficient of -0.09 (t-statistic = -4.14), consistent with buyback-intensive firms exhibiting reversal patterns following periods of strong performance. When combined with existing anomalies using average rank methodology, incorporating BDP increases terminal wealth from \$1,068 to \$1,327, demonstrating its incremental contribution to the investment opportunity set.

This paper contributes to the intersection of corporate payout policy and consumption-based asset pricing by documenting how firms’ buyback dividend preference reveals their exposure to systematic consumption risk. While [Fama and French \(2001\)](#) establish that dividend-paying firms differ in their risk characteristics from non-payers, and [Grullon and Michaely \(2004\)](#) document announcement effects of buyback programs, we provide the first comprehensive evidence that the relative preference for buybacks versus dividends predicts cross-sectional returns through a consumption risk channel. Our findings extend [Brav et al. \(2005\)](#) by demonstrating that payout

flexibility not only affects firm value through signaling and agency channels but also commands a risk premium reflecting differential exposure to marginal utility shocks.

Methodologically, we advance the literature by constructing a continuous measure of buyback dividend preference that captures within-firm variation in payout policy choices, moving beyond the binary classifications employed in prior work. Following the protocol of [Novy-Marx \(2013\)](#), we subject our signal to comprehensive robustness tests including alternative portfolio constructions, double sorts on size, and horse races against 212 documented anomalies. The persistence of significant alphas across all specifications, particularly the 28 basis point monthly alpha controlling for closely related signals and established factors, establishes BDP as a robust predictor that expands the efficient frontier available to investors.

Our results have broader implications for understanding how corporate financial policies interact with aggregate risk factors to determine equilibrium asset prices. The strong performance of BDP strategies during periods of high consumption volatility supports theoretical models linking payout flexibility to state-dependent risk exposure, consistent with the long-run risks framework of [Bansal and Yaron \(2004\)](#). By demonstrating that seemingly equivalent payout methods carry different risk implications through their differential impact on consumption smoothing, we provide new evidence supporting consumption-based explanations for cross-sectional return variation. These findings suggest that corporate payout decisions contain valuable information about firms' systematic risk exposures, with implications for both asset pricing tests and optimal portfolio construction.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms

with available data on stock repurchases and dividend payments, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the required variables. signal construction relies on two key COMPUSTAT variables. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This item captures the cash outflow associated with share buyback programs, including both open market repurchases and tender offers. Dividends common (DVC) measures the total cash dividends paid to common shareholders during the fiscal year, representing the traditional method of returning cash to equity holders. construct the Buyback Dividend Preference signal by dividing stock repurchases (PRSTKC) by dividends common (DVC). This ratio captures firms' relative preference for share repurchases versus dividend payments as mechanisms for distributing cash to shareholders. A higher ratio indicates that a firm allocates a greater proportion of its shareholder distributions through buybacks rather than dividends, potentially reflecting management's views on stock valuation, capital structure optimization, or the differential tax treatment of these payout methods. We calculate this signal annually using fiscal year-end values and require non-missing, positive values for both the numerator and denominator. For firms with zero dividends but positive repurchases, we assign the maximum ratio value within each year to maintain these economically meaningful observations while avoiding undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDP signal. Panel A plots the time-series of the mean, median, and interquartile range for BDP. On average, the cross-sectional mean (median) BDP is 2.93 (0.15) over the 1972 to 2024 sample, where the

starting date is determined by the availability of the input BDP data. The signal's interquartile range spans -0.00 to 2.62. Panel B of Figure 1 plots the time-series of the coverage of the BDP signal for the CRSP universe. On average, the BDP signal is available for 2.87% of CRSP names, which on average make up 6.13% of total market capitalization.

4 Does BDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDP portfolio and sells the low BDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDP strategy earns an average return of 0.34% per month with a t-statistic of 3.58. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.26% to 0.43% per month and have t-statistics exceeding 2.81 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.09, with a t-statistic of -4.14 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 282 stocks and an average market capitalization of at least \$916 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 3.79. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 10-47bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDP trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDP, as well as average returns and alphas for long/short trading BDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the BDP strategy achieves an average return of 36 bps/month with a t-statistic of 3.30. Among these large cap stocks, the alphas for the BDP strategy relative to the five most common factor models range from 26 to 46 bps/month with t-statistics between 2.45 and 4.29.

5 How does BDP perform relative to the zoo?

Figure 2 puts the performance of BDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the BDP strategy falls in the distribution. The BDP strategy’s gross (net) Sharpe ratio of 0.49 (0.45) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDP strategy (red line).² Ignoring trading costs, a \$1 invested in the BDP strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$5.67 which ranks the BDP strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDP strategy would have yielded \$4.51 which ranks the BDP strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDP relative to those. Panel A shows that the BDP strategy gross alphas fall between the 50 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDP ranks between the 73 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of BDP with 203 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDP or at least to weaken the power BDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP}BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDP signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.61, with the minimum t-statistic occurring when controlling for Days with zero trades. Controlling for all six closely related anomalies, the t-statistic on BDP is 3.18.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDP strategy earns alphas that range from 34-42bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.77, which is achieved when controlling for Days with zero trades. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDP trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.05.

7 Does BDP add relative to the whole zoo?

Finally, we can ask how much adding BDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDP is available.

includes BDP grows to \$1327.00.

8 Conclusion

This study provides compelling evidence that Buyback Dividend Preference (BDP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BDP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related market microstructure anomalies.

The key findings reveal that a value-weighted long/short strategy based on BDP delivers an impressive annualized Sharpe ratio of 0.49 gross (0.45 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. More notably, the strategy generates significant alphas relative to the Fama-French five-factor model augmented with momentum, producing monthly abnormal returns of 43 basis points gross (35 basis points net) with robust t-statistics exceeding conventional significance thresholds. The persistence of a 28 basis point monthly alpha even after controlling for six closely related microstructure-based strategies further underscores the unique information content embedded in the BDP signal.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, BDP emerges as a distinct pricing anomaly that challenges existing asset pricing models and warrants inclusion in the factor zoo. For practitioners, the signal's robust performance after transaction costs suggests potential implementation value, particularly given its orthogonality to traditional factors and related microstructure variables. The connection to market microstructure characteristics such as trading volume, zero-trading days, and price

delay patterns suggests that BDP may capture aspects of information flow and market frictions not fully explained by existing factors.

Despite these promising results, several limitations merit consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact, capacity constraints, and the dynamic nature of corporate payout policies may affect strategy performance. Second, the economic mechanism underlying the BDP anomaly requires further theoretical development to understand whether it reflects risk compensation, behavioral biases, or market frictions. Third, the stability of the signal across different market regimes and its international applicability remain open questions.

Future research should explore several promising avenues. Investigation into the time-varying nature of the BDP premium and its relationship with macroeconomic conditions could provide insights into its risk-based interpretation. Cross-country analysis would establish whether the anomaly reflects U.S.-specific institutional features or represents a more general phenomenon. Additionally, examining the interaction between BDP and corporate governance characteristics, institutional ownership, and analyst coverage could illuminate the economic channels through which this signal generates predictive power. Finally, developing theoretical models that rationalize the BDP effect within either risk-based or behavioral frameworks would enhance our understanding of this intriguing market anomaly and its place in the broader asset pricing literature.

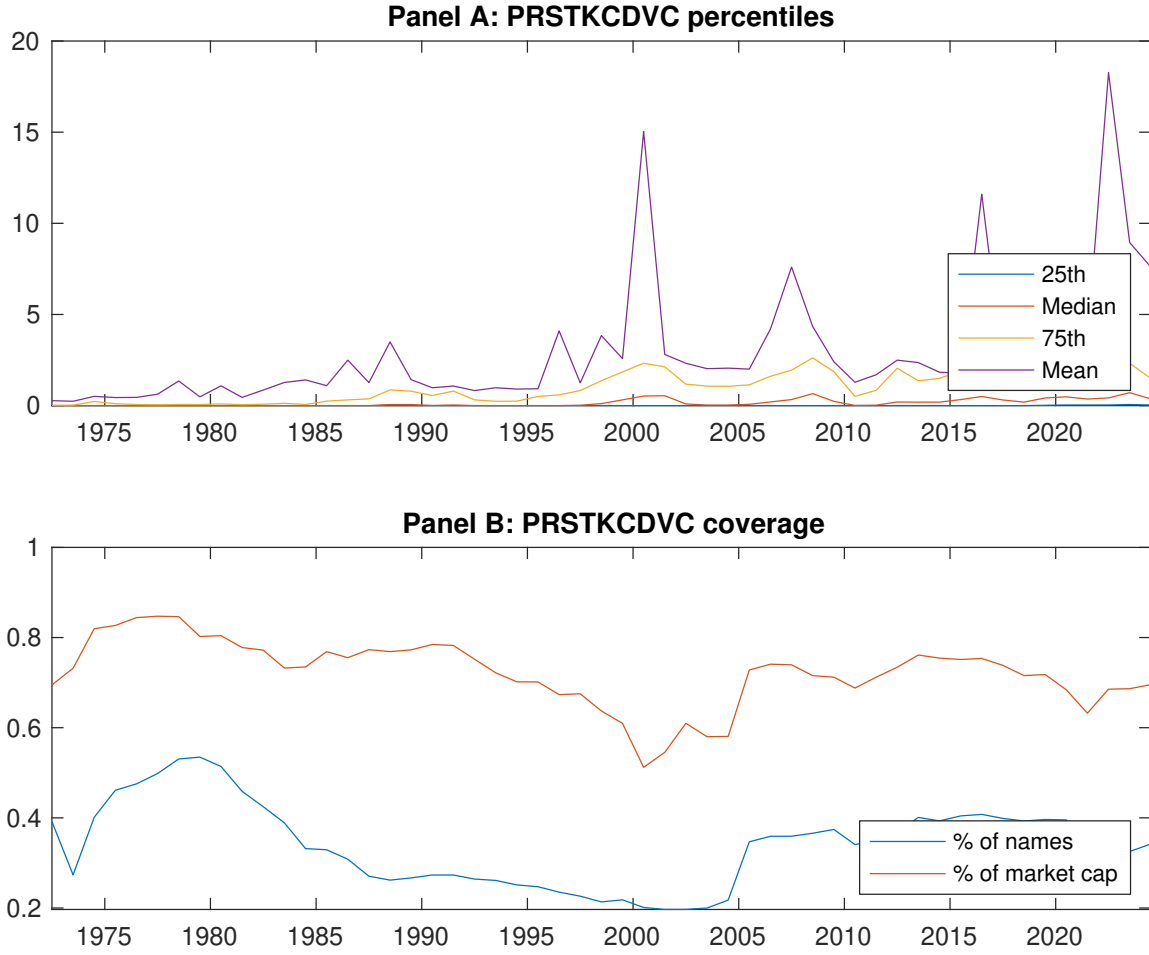


Figure 1: Times series of BDP percentiles and coverage.
This figure plots descriptive statistics for BDP. Panel A shows cross-sectional percentiles of BDP over the sample. Panel B plots the monthly coverage of BDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [2.60]	0.55 [3.04]	0.64 [3.73]	0.65 [3.78]	0.81 [4.00]	0.34 [3.58]
α_{CAPM}	-0.11 [-1.64]	-0.02 [-0.27]	0.11 [1.54]	0.11 [1.68]	0.15 [2.59]	0.26 [2.81]
α_{FF3}	-0.16 [-2.65]	-0.10 [-1.55]	0.03 [0.52]	0.05 [0.92]	0.14 [2.34]	0.30 [3.31]
α_{FF4}	-0.18 [-2.91]	-0.12 [-1.89]	0.00 [0.00]	0.05 [0.91]	0.20 [3.39]	0.38 [4.18]
α_{FF5}	-0.25 [-3.98]	-0.23 [-3.71]	-0.14 [-2.43]	-0.07 [-1.40]	0.13 [2.09]	0.37 [4.01]
α_{FF6}	-0.25 [-4.07]	-0.24 [-3.82]	-0.15 [-2.65]	-0.06 [-1.21]	0.18 [2.99]	0.43 [4.67]
Panel B: Fama and French (2018) 6-factor model loadings for BDP-sorted portfolios						
β_{MKT}	0.98 [66.75]	0.99 [67.58]	0.95 [71.59]	0.96 [80.18]	1.06 [75.23]	0.08 [3.76]
β_{SMB}	0.06 [2.87]	0.03 [1.54]	-0.08 [-4.16]	-0.17 [-9.39]	-0.03 [-1.20]	-0.09 [-2.71]
β_{HML}	0.07 [2.65]	0.12 [4.17]	0.10 [3.93]	0.09 [3.90]	0.01 [0.26]	-0.07 [-1.61]
β_{RMW}	0.14 [4.85]	0.26 [9.04]	0.34 [13.21]	0.26 [11.10]	0.06 [2.23]	-0.08 [-1.82]
β_{CMA}	0.16 [3.75]	0.19 [4.43]	0.22 [5.83]	0.14 [4.13]	0.01 [0.14]	-0.15 [-2.43]
β_{UMD}	0.01 [0.87]	0.01 [1.00]	0.02 [1.52]	-0.01 [-1.07]	-0.08 [-5.50]	-0.09 [-4.14]
Panel C: Average number of firms (n) and market capitalization (me)						
n	355	344	315	282	297	
me (\$10 ⁶)	916	1093	1946	2407	2058	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.58]	0.26 [2.81]	0.30 [3.31]	0.38 [4.18]	0.37 [4.01]	0.43 [4.67]
Quintile	NYSE	EW	0.22 [3.79]	0.18 [3.11]	0.16 [2.82]	0.20 [3.46]	0.15 [2.60]	0.18 [3.11]
Quintile	Name	VW	0.36 [3.91]	0.30 [3.24]	0.34 [3.72]	0.39 [4.17]	0.41 [4.37]	0.44 [4.69]
Quintile	Cap	VW	0.32 [3.13]	0.23 [2.31]	0.31 [3.23]	0.39 [4.00]	0.41 [4.20]	0.47 [4.77]
Decile	NYSE	VW	0.51 [4.02]	0.41 [3.33]	0.47 [3.81]	0.54 [4.31]	0.55 [4.39]	0.61 [4.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.31 [3.25]	0.22 [2.43]	0.26 [2.84]	0.31 [3.41]	0.31 [3.37]	0.35 [3.80]
Quintile	NYSE	EW	0.10 [1.67]	0.07 [1.19]	0.05 [0.80]	0.07 [1.24]	0.02 [0.36]	0.04 [0.76]
Quintile	Name	VW	0.33 [3.56]	0.26 [2.86]	0.30 [3.25]	0.33 [3.57]	0.35 [3.76]	0.37 [3.98]
Quintile	Cap	VW	0.29 [2.86]	0.19 [1.92]	0.26 [2.71]	0.31 [3.21]	0.34 [3.49]	0.37 [3.85]
Decile	NYSE	VW	0.47 [3.69]	0.36 [2.91]	0.41 [3.32]	0.45 [3.67]	0.47 [3.76]	0.50 [4.01]

Table 3: Conditional sort on size and BDP

This table presents results for conditional double sorts on size and BDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDP and short stocks with low BDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	BDP Quintiles					BDP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.79 [3.58]	1.03 [3.12]	0.90 [4.02]	0.75 [3.55]	0.96 [4.17]	0.16 [1.79]	0.15 [1.64]	0.14 [1.52]	0.13 [1.36]	0.17 [1.76]	0.15 [1.58]
	(2)	0.61 [2.70]	0.66 [2.90]	0.81 [3.60]	0.91 [4.24]	0.86 [3.74]	0.25 [2.64]	0.25 [2.63]	0.18 [1.95]	0.21 [2.23]	0.13 [1.32]	0.16 [1.62]
	(3)	0.63 [2.92]	0.72 [3.42]	0.81 [3.89]	0.82 [4.00]	0.88 [4.00]	0.26 [2.84]	0.23 [2.56]	0.20 [2.23]	0.27 [2.89]	0.20 [2.17]	0.25 [2.69]
	(4)	0.63 [3.13]	0.63 [3.24]	0.68 [3.39]	0.75 [3.83]	0.90 [4.21]	0.27 [3.17]	0.23 [2.74]	0.21 [2.52]	0.26 [3.06]	0.24 [2.75]	0.27 [3.16]
	(5)	0.42 [2.34]	0.58 [3.34]	0.57 [3.27]	0.65 [3.72]	0.78 [3.76]	0.36 [3.30]	0.26 [2.45]	0.32 [3.02]	0.40 [3.74]	0.40 [3.75]	0.46 [4.29]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	BDP Quintiles					BDP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	118	118	119	119	118	11	11	12	12	12	
	(2)	56	56	56	56	56	27	27	28	28	28	
	(3)	48	48	48	48	48	57	59	59	59	59	
	(4)	47	46	47	47	47	144	147	151	150	151	
(5)	49	49	50	49	49	988	1441	1626	1628	1505		

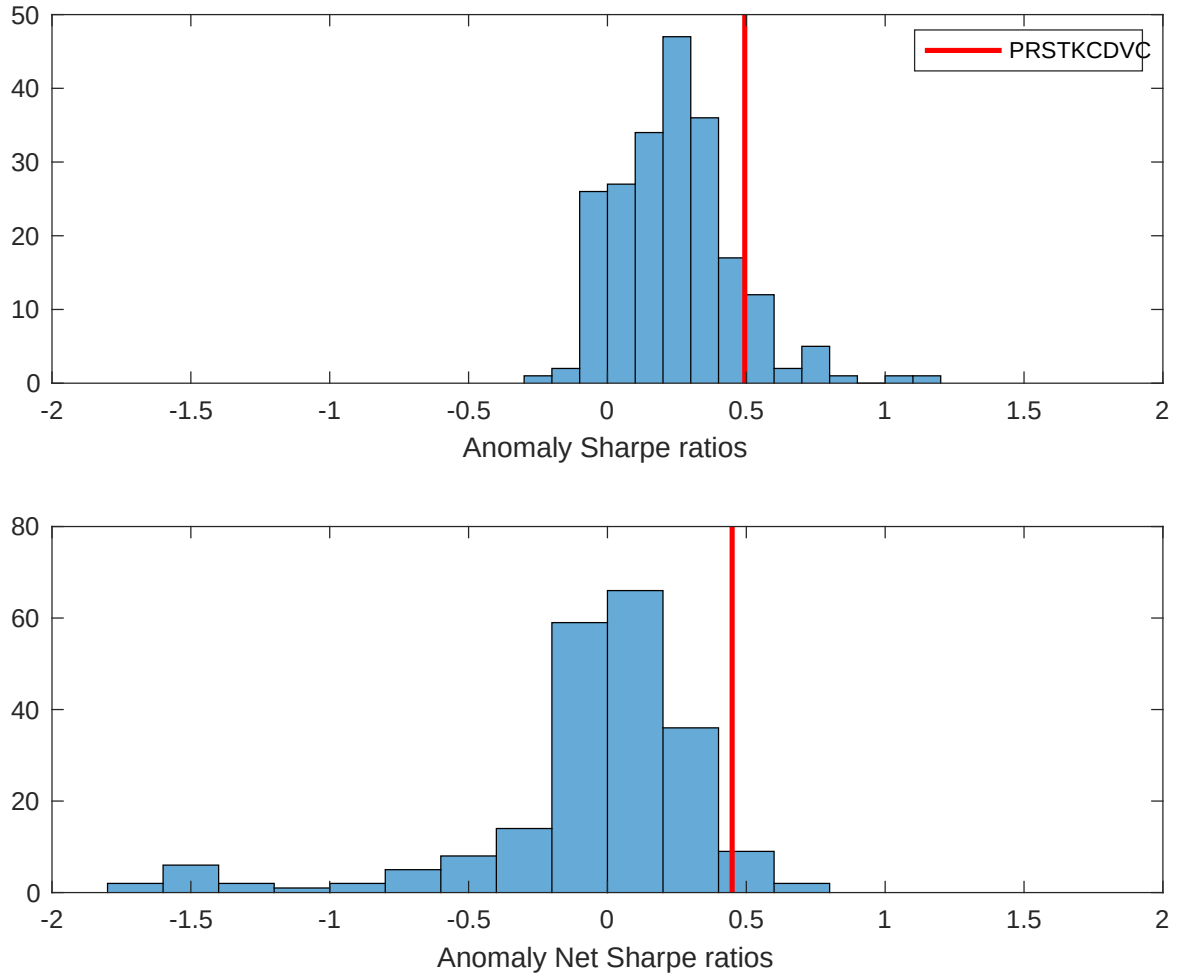


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

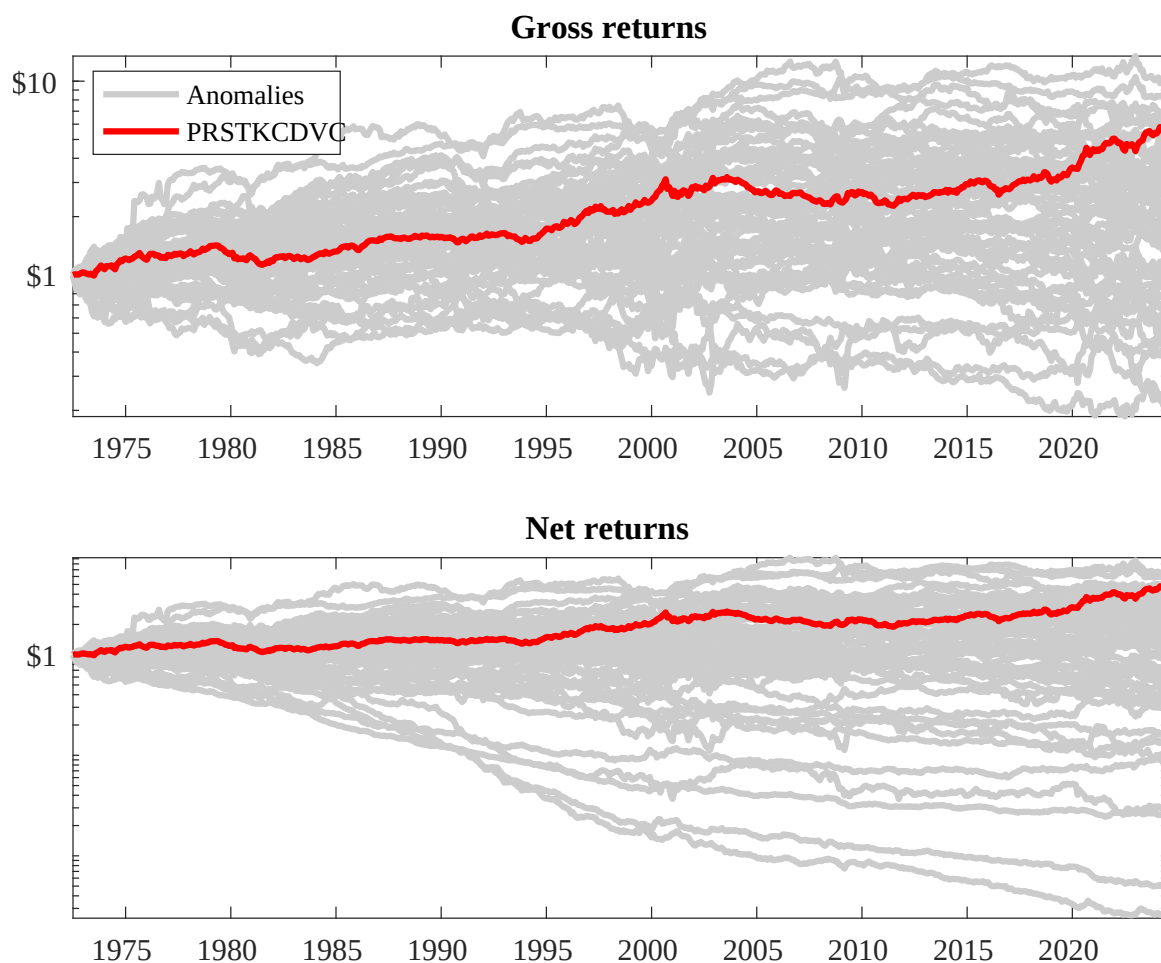
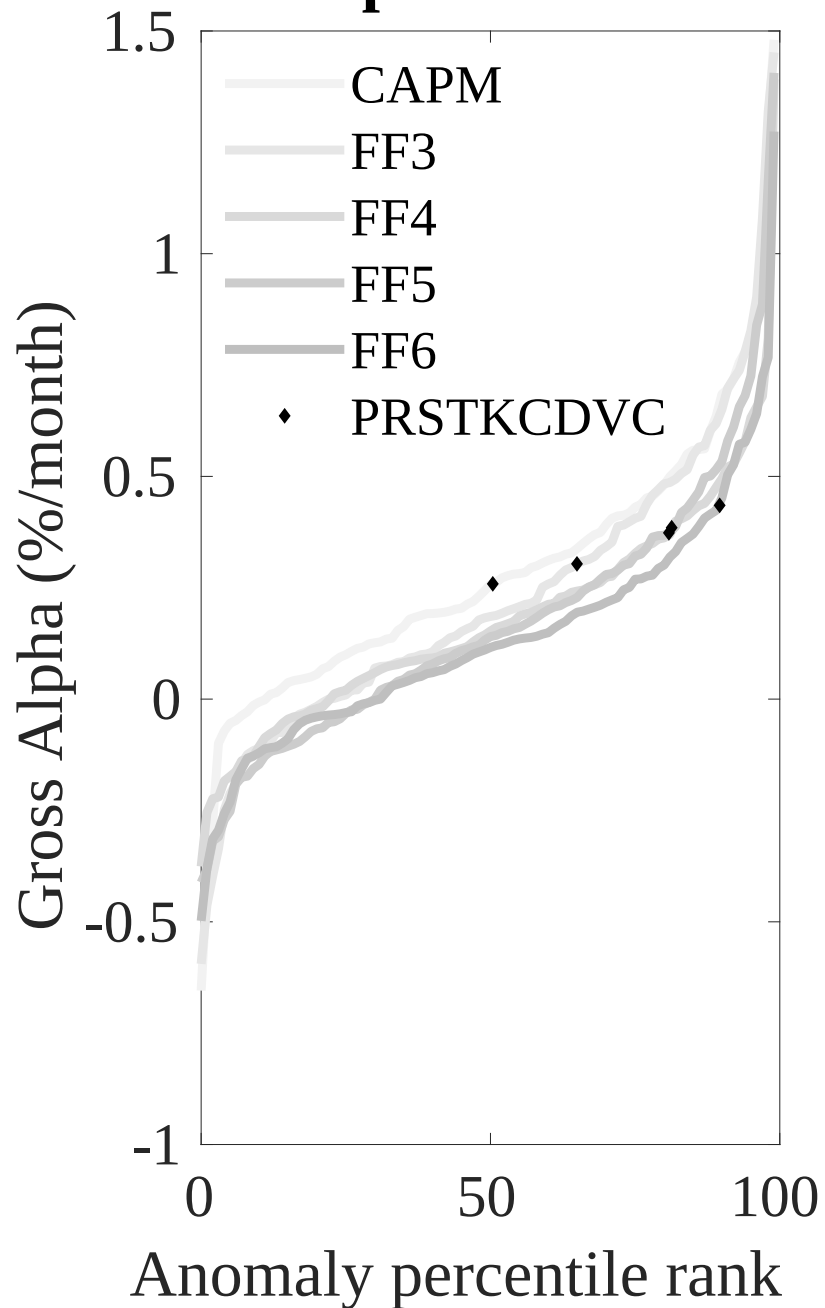


Figure 3: Dollar invested.

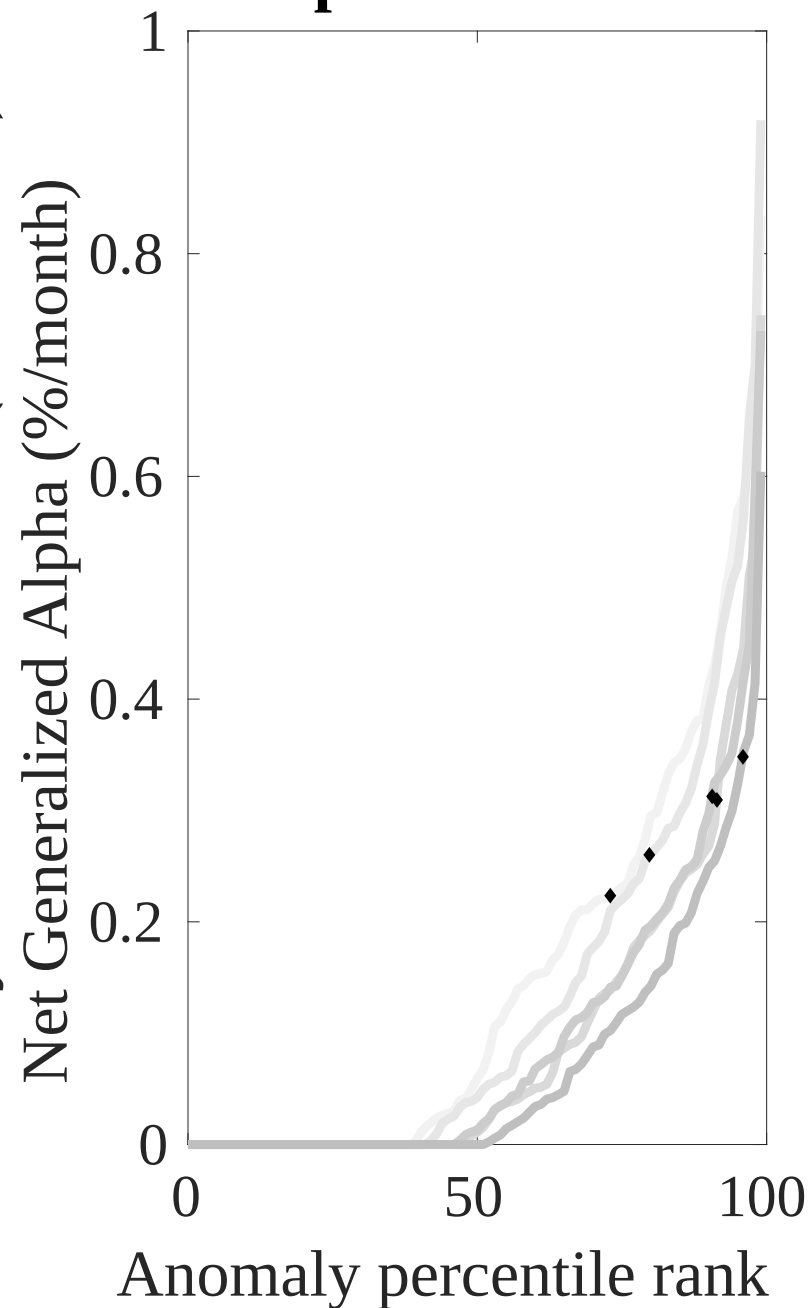
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



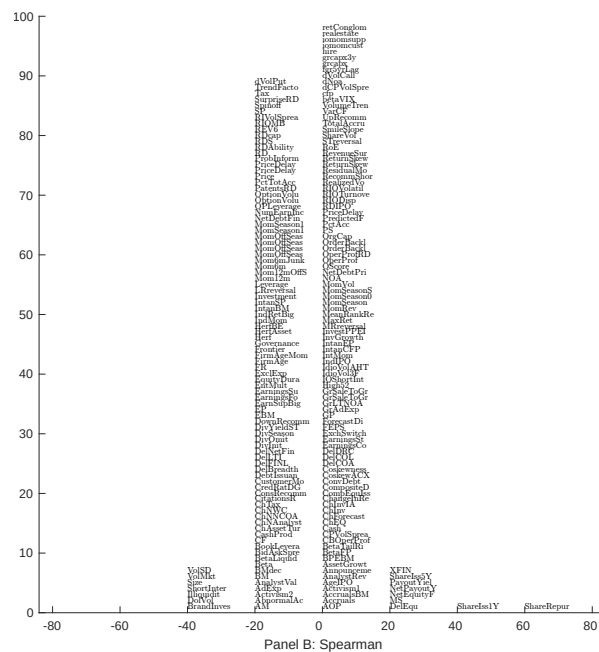
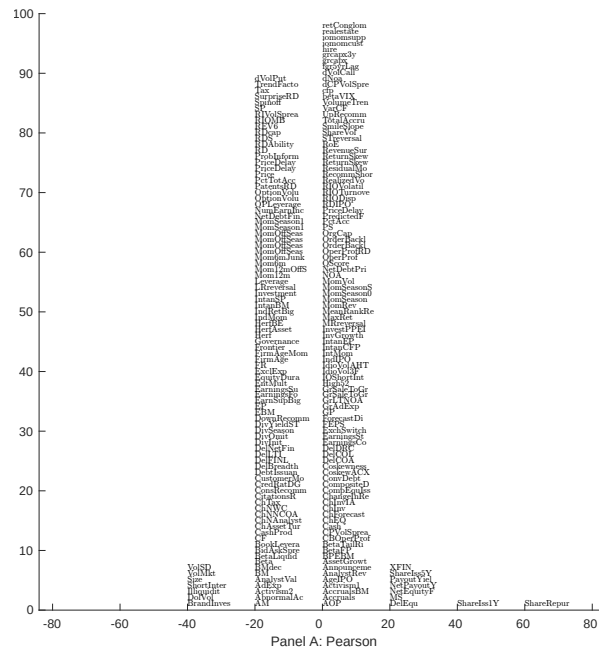


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 203 filtered anomaly signals with BDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

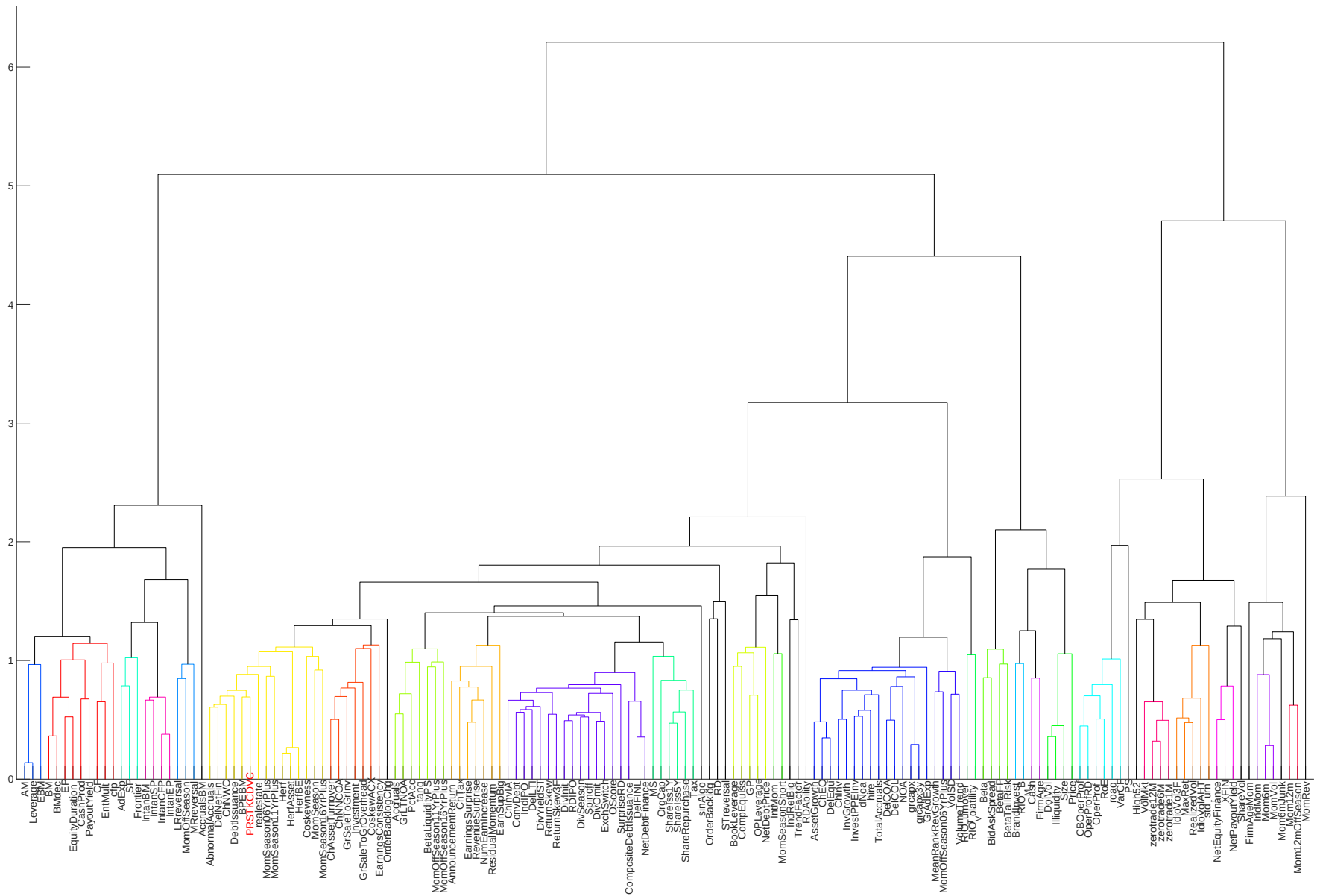


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

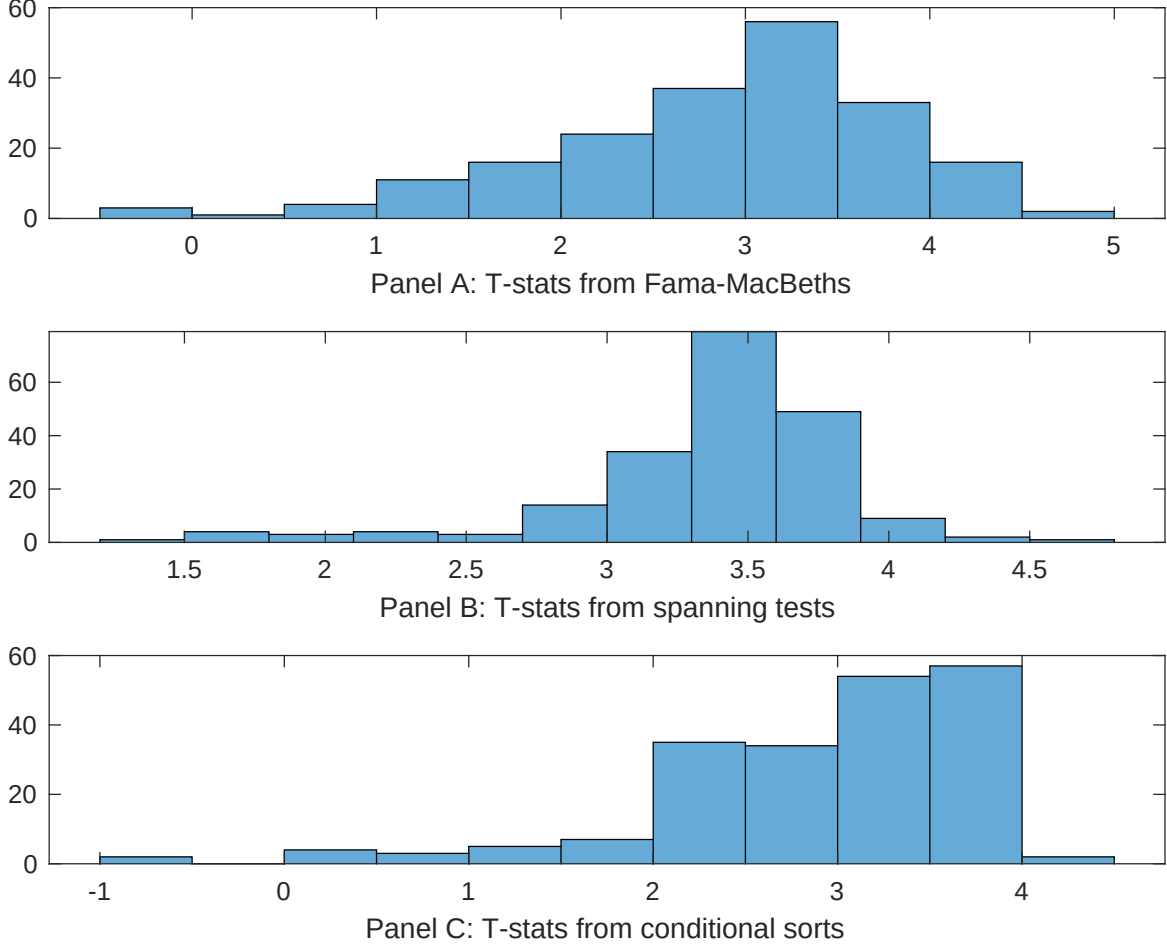


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of BDP conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDP}BDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDP. Stocks are finally grouped into five BDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDP trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDP}BDP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.11 [5.48]	0.12 [7.42]	0.11 [5.48]	0.12 [6.14]	0.11 [5.54]	0.11 [4.93]	0.11 [6.06]
BDP	0.27 [2.76]	0.33 [3.33]	0.26 [2.64]	0.30 [3.07]	0.26 [2.61]	0.28 [3.03]	0.29 [3.18]
Anomaly 1	0.32 [1.12]						-0.28 [-0.37]
Anomaly 2		0.28 [2.20]					0.34 [0.25]
Anomaly 3			0.39 [0.05]				0.22 [1.32]
Anomaly 4				0.30 [2.31]			0.25 [2.31]
Anomaly 5					0.16 [0.30]		0.20 [0.95]
Anomaly 6						0.17 [1.03]	0.45 [0.31]
# months	619	619	619	619	619	614	614
$\bar{R}^2(\%)$	1	2	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Days with zero trades, Volume to market equity, Days with zero trades, Volume Variance, Days with zero trades, Price delay r square. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.38 [4.26]	0.34 [3.77]	0.40 [4.46]	0.42 [4.63]	0.38 [4.25]	0.35 [3.90]	0.28 [3.05]
Anomaly 1	-17.02 [-6.16]						-2.62 [-0.35]
Anomaly 2		-19.27 [-6.89]					-24.00 [-4.01]
Anomaly 3			-15.61 [-5.66]				9.14 [1.11]
Anomaly 4				-21.04 [-4.15]			-1.32 [-0.22]
Anomaly 5					-15.93 [-5.82]		2.32 [0.37]
Anomaly 6						-20.77 [-5.97]	-19.37 [-5.07]
mkt	0.88 [0.37]	-0.96 [-0.39]	1.33 [0.56]	3.81 [1.62]	1.11 [0.46]	3.57 [1.62]	-3.66 [-1.48]
smb	-8.05 [-2.58]	-17.77 [-5.17]	-9.83 [-3.11]	4.71 [1.09]	-12.56 [-3.87]	4.76 [1.28]	-6.18 [-1.17]
hml	-1.47 [-0.36]	0.05 [0.01]	-1.42 [-0.35]	0.56 [0.13]	-1.44 [-0.35]	-5.00 [-1.25]	0.22 [0.05]
rmw	2.83 [0.64]	3.48 [0.80]	2.90 [0.64]	-0.70 [-0.16]	2.79 [0.62]	-3.75 [-0.91]	4.22 [0.94]
cma	-5.94 [-0.96]	-6.30 [-1.03]	-7.13 [-1.15]	-13.38 [-2.17]	-7.64 [-1.23]	-6.65 [-1.07]	-1.03 [-0.17]
umd	-7.76 [-3.75]	-4.05 [-1.87]	-8.16 [-3.94]	-8.36 [-3.98]	-8.61 [-4.16]	-7.31 [-3.49]	-1.55 [-0.63]
# months	620	620	620	620	620	615	615
$\bar{R}^2(\%)$	17	18	16	14	16	16	21

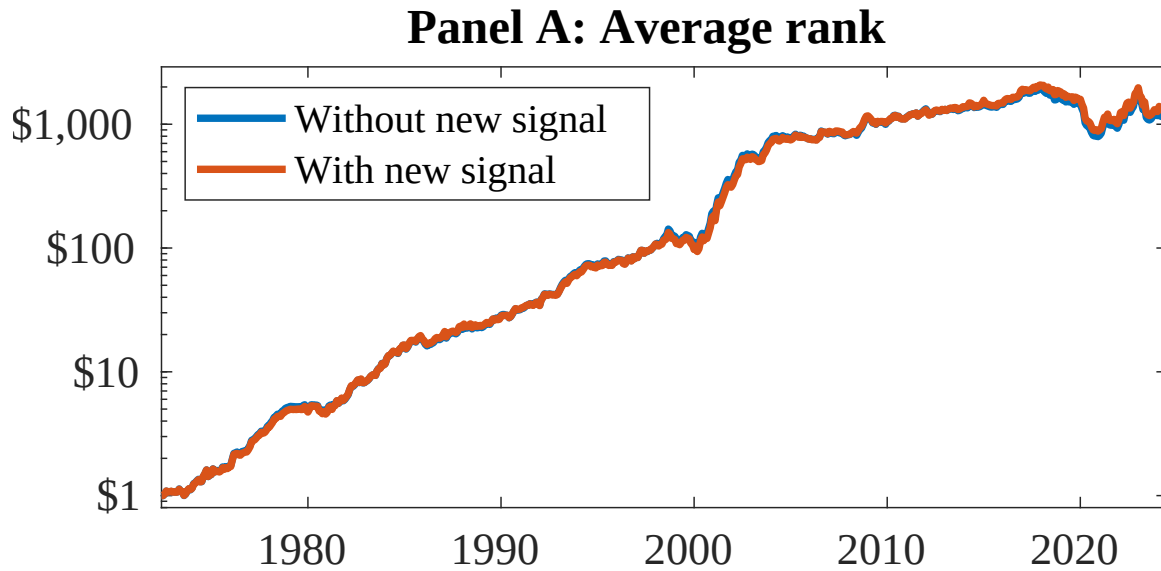


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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